Tirtharaj Dash

tirtharaj [at] goa.bits-pilani.ac.in ◆ tirtharajdash.github.io ◆ G-scholar:1ZcwKZEAAAAJ

Research Interests

Main: Developing neurosymbolic deep learning models (graph neural networks, LLMs, sparse and logic-structured networks) that incorporate scientific domain knowledge, with an emphasis on interpretability and data efficiency; applying them to problems in the life sciences, including drug discovery, gene regulation, genomic sequence analysis and multi-omics cancer analysis.

Keywords: Neurosymbolic AI, Deep Learning, AI for Science

Experience

BITS Pilani, K K Birla Goa Campus Dec 2025 – Present
Assistant Professor (Grade I), Dept. of Computer Science
Research: Neurosymbolic AI, Deep Learning, AI for Science (Drug Discovery, Molecular Biology)

University of Cambridge, UK Oct 2023 – Dec 2025
Research Associate (Postdoc), Bornelöv Lab, with Susanne Bornelöv
Research: AI in genomics: explainable and foundation models for gene regulation

University of California, San Diego Oct 2022 – Sep 2023
Postdoctoral Scholar, Boolean Lab, with Debashis Sahoo
Research: AI in oncology: macrophage polarisation and cancer biomarker discovery

BITS Pilani, K K Birla Goa Campus Aug 2015 – Aug 2022
Assistant Professor (Grade II), Dept. of Computer Science; APPCAIR (from Jun 2020)
Research: AI and Machine Learning; industry collaborations (TCS, Reflexis)

National Institute of Science & Technology Berhampur Jun 2014 – Aug 2015
Assistant Professor, School of Computer Science
Research: Protein sequence analysis, optimisation algorithms, and machine learning

Earlier (summer research internships):
NIST Berhampur, with Prabhat K. Sahu: protein structure optimisation (2016);
ISI Kolkata, with Nikhil R. Pal: self-organising maps (2015)

Education

Birla Institute of Technology and Science Pilani, India Jan 2017 – Jul 2022
Ph.D. in Computer Science (Thesis submitted: May 13, 2022, defended: Jul 19, 2022)
Thesis: *Inclusion of Symbolic Domain-Knowledge into Deep Neural Networks* ▹www
Supervisor: Ashwin Srinivasan (co-supervisor: Sukanta Mondal)

Veer Surendra Sai University of Technology Burla, India Jul 2012 – Jun 2014
M.Tech. in Computer Science & Engineering (*GPA: 9.78/10; University Silver Medalist*)
Thesis: *Pattern Recognition using Soft Computing Approaches* ▹www
Supervisor: H.S. Behera

National Institute of Science and Technology, Berhampur, India Aug 2008 – Jul 2012
B.Tech. in Information Technology (*GPA: 8.91/10; Institute Silver Medalist*)
Thesis: *Designing Parallel Algorithms for Solving Ordinary Differential Equations*
Supervisor: Motahar Reza

Selected Honours & Awards

Selected as a **Postdoctoral Affiliate** of **Trinity College, Cambridge**	2024
Best PhD Thesis Award from BITS Pilani, Goa campus (one of two recipients)	2023
Best Short Paper Award from the ACM in ACMSE	2022
ICML 2021 Workshop on Computational Biology Fellowship	2021
AWSAR Award 2019 from DST, Govt. of India (Country Rank: **9** in Best-100 Category)	2020
Best Student Research Paper Award from the Machine Learning Journal in ILP	2018
Travel Grant from EurAI to attend ACAI-2018, Ferrara, Italy	2018
IASc-INSA-NASI Summer Research Fellowship	2015

Grants

1. Indian Ocean Region Seaport and Vessel Traffic Atlas Project; Funded by: TLC, BITS Pilani, Goa Campus; INR 140,000; Period: Oct 2021 to Mar 2022; Role: Co-PI.
2. DataLab Project: TCS-BITS Collaboration; Funded by: TCS Ltd., India; INR 6,289,000; Period: Jan 2019 to Dec 2021; Role: PI (1 year), Co-PI (2 years).
3. CoLab Project: Reflexis-BITS Collaboration; Funded by: Reflexis Systems, USA; INR 2,500,000; Period: Jul 2019 to Dec 2019; Role: Co-PI.

Refereed Journal Publications

1. A. Srinivasan, T. Dash, A. Baskar, M. Bain, S.K. Dey, M. Banerjee, "Symbolic Neural Generation with Applications to Lead Discovery in Drug Design", *Machine Learning*, **2026**. **MLJ**
2. T. Dash, S. Bornelöv, "Predicting gene expression using millions of yeast promoters reveals *cis*-regulatory logic", *Bioinformatics Advances*, **2025**. **BioAdv**
3. M. Mahajan, S. Dhabalia, T. Dash, A. Sarkar, S. Mondal, "A comprehensive multi-omics study reveals potential prognostic and diagnostic biomarkers for colorectal cancer", *International Journal of Biological Macromolecules*, **2025**. **IJBIOMAC**
4. A. Srinivasan, A. Baskar, T. Dash, D. Shah, "Composition of Relational Features with an Application to Explaining Black-Box Predictors", *Machine Learning*, **2023**. **MLJ**
5. T. Dash, S. Chitlangia, A. Ahuja, A. Srinivasan, "A Review of Some Techniques for Inclusion of Domain-Knowledge into Deep Neural Networks", *Scientific Reports*, **2022**. **SCI. REP**
6. I. Olier, O.I. Orhobor, T. Dash, A. Davis, L.N. Soldatova, J. Vanschoren, R.D. King, "Transformational machine learning: Learning how to learn from many related scientific problems", *Proceedings of the National Academy of Sciences*, **2021**. (news) **PNAS**
7. T. Dash, A. Srinivasan, A. Baskar, "Inclusion of domain-knowledge into GNNs using mode-directed inverse entailment", *Machine Learning*, **2021**. **MLJ**
8. T. Dash, A. Srinivasan, L. Vig, "Incorporating symbolic domain knowledge into graph neural networks", *Machine Learning*, **2021**. **MLJ**
9. R. Kaushik, S. Jain, S. Jain, T. Dash, "Performance evaluation of deep neural networks for forecasting time-series with multiple structural breaks and high volatility", *CAAI Transactions on Intelligence Technology*, **2021**. **CAAI TIT**
10. T. Dash, S.N. Dambekodi, P.N. Reddy, A. Abraham, "Adversarial neural networks for playing hide-and-search board game Scotland Yard", *Neural Computing and Applications*, **2018**. **NCAA**

11. T. Dash, H.S. Behera, "A comprehensive study on evolutionary algorithm-based multilayer perceptron for real-world data classification under uncertainty", *Expert Systems*, **2018**. **EXSY**

12. P.P. Pai, T. Dash, S. Mondal, "Sequence-based discrimination of protein-RNA interacting residues using a probabilistic approach", *Journal of Theoretical Biology*, **2017**. **JTB**

13. T. Dash, "A study on intrusion detection using neural networks trained with evolutionary algorithms", *Soft Computing*, **2015**. **SOCO**

14. T. Dash, "Automatic navigation of wall following mobile robot using Adaptive Resonance Theory of Type-1", *Biologically Inspired Cognitive Architectures*, **2015**. **BICA**

15. T. Dash, P.K. Sahu, "Gradient Gravitational Search: An Efficient Metaheuristic Algorithm for Global Optimization", *Journal of Computational Chemistry*, **2015**. **JCC**

Refereed Conference and Workshop Publications

1. A. Rathore, T. Dash, L. Vig, A. Srinivasan, "Robustness of Diffusion and Autoregressive Language Models to CheckList-Style Perturbations", *Grounding Language Models: Learning Faithfully and Efficiently @ EMNLP 2026*, **2026**. **GroundLM@EMNLP**

2. T. Dash, "BIRDNet: Mining and Encoding Boolean Implication Knowledge Graphs as Interpretable Deep Neural Networks", *ACM International Conference on Information and Knowledge Management*, **2026**. **CIKM**

3. S.B. Brahmavar, A. Srinivasan, T. Dash, S.R. Krishnan, L. Vig, A. Roy, R. Aduri, "Generating Novel Leads for Drug Discovery using LLMs with Logical Feedback", *AAAI Conference on Artificial Intelligence*, **2024**. **AAAI**

4. S.B. Brahmavar, R. Rajesh, T. Dash, L. Vig, T.T. Verlekar, M.M. Hasan, T. Khan, E. Meijering, A. Srinivasan, "IKD+: Reliable Low Complexity Deep Models for Retinopathy Classification", *International Conference on Image Processing*, **2023**. **ICIP**

5. S.R. Chitnis, S. Liu, T. Dash, T.T. Verlekar, A. Di Ieva, S. Berkovsky, L. Vig, A. Srinivasan, "Domain-Specific Pretraining Improves Confidence in Whole Slide Image Classification", *Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, **2023**. **EMBC**

6. R. Patra, R. Hebbalaguppe, T. Dash, G. Shroff, L. Vig, "Calibrating Deep Neural Networks using Explicit Regularisation and Dynamic Data Pruning", *IEEE/CVF Winter Conference on Applications of Computer Vision*, **2023**. (Spotlight Paper) **WACV**

7. V. Shah, A. Sharma, G. Shroff, L. Vig, T. Dash, A. Srinivasan, "Knowledge-based Analogical Reasoning in Neuro-symbolic Latent Spaces", *International Workshop on Neural-Symbolic Learning and Reasoning*, **2022**. **NESY**

8. G. Chhablani, A. Sharma, H. Pandey, T. Dash, "Superpixel-based Knowledge Infusion in Deep Neural Networks for Image Classification", *ACM Southeast Regional Conference*, **2022**. (Best Short Paper Award) **ACMSE**

9. A. Sonwane, G. Shroff, L. Vig, A. Srinivasan, T. Dash, "Solving Visual Analogies Using Neural Algorithmic Reasoning (Student Abstract)", *AAAI Conf on Artificial Intelligence*, **2022**. **AAAI**

10. A. Lalwani, A. Saraiya, A. Singh, A. Jain, T. Dash, "Machine Learning in Sports: A Case Study on Using Explainable Models for Predicting Outcomes of Volleyball Matches", *International Conference on Sports Engineering*, **2022**. **ICSE**

11. S. Chitlangia, A. Sonwane, T. Dash, L. Vig, A. Srinivasan, G. Shroff, "Using Program Synthesis and Inductive Logic Programming to solve Bongard Problems", *International Workshop on Approaches and Applications of Inductive Programming*, **2021**. **AAIP**

12. T. Dash, A. Srinivasan, L. Vig, A. Roy, "Using Domain-Knowledge to Assist Lead Discovery in Early-Stage Drug Design", *Intl Conf on Inductive Logic Programming*, **2021**. **ILP**

13. H. Shah, A. Vaswani, T. Dash, R. Hebbalaguppe, A. Srinivasan, "Empirical Study of Data-Free Iterative Knowledge Distillation", *Intl Conf on Artificial Neural Netw*, **2021**. **ICANN**

14. A. Sharma, H. Pandey, G. Chhablani, Y. Bhartia, T. Dash, "LRG at SemEval-2021 Task 4: Improving Reading Comprehension with Abstract Words using Augmentation, Linguistic Features and Voting", *International Workshop on Semantic Evaluation*, **2021**. **SEMEVAL**

15. K. Mahajan, M. Sharma, L. Vig, R. Khincha, S. Krishnan, A. Niranjan, T. Dash, A. Srinivasan, G. Shroff, "CovidDiagnosis: Deep Diagnosis of Covid-19 Patients using Chest X-rays", *International Workshop on Thoracic Image Analysis*, **2020**. **TIA@MICCAI**

16. S. Krishnan, R. Khincha, L. Vig, T. Dash, A. Srinivasan, "A Case Study of Transfer of Lesion-Knowledge", *International Workshop on Medical Image Learning with Less Labels and Imperfect Data*, **2020**. **MIL3ID@MICCAI**

17. S. Yalburgi, T. Dash, R. Hebbalaguppe, S. Hegde, A. Srinivasan, "An Empirical Study of Iterative Knowledge Distillation for Neural Network Compression", *European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning*, **2020**. **ESANN**

18. T. Dash, A. Srinivasan, R.S. Joshi, A. Baskar, "Discrete Stochastic Search and Its Application to Feature-Selection for Deep Relational Machines", *Intl Conf on Artif. Neural Netw*, **2019**. **ICANN**

19. T. Dash, A. Srinivasan, L. Vig, O.I. Orhobor, R.D. King, "Large-Scale Assessment of Deep Relational Machines", *Intl Conf on Inductive Logic Programming*, **2018**. (Best Student Paper Award) **ILP**

20. P.S.M. Saladi, T. Dash, "Genetic Algorithm-Based Oversampling Technique to Learn from Imbalanced Data", *Intl Conf on Soft Computing for Problem Solving*, **2017**. **SocProS**

21. S. Iyer, S. Chaturvedi, T. Dash, "Image Captioning-Based Image Search Engine: An Alternative to Retrieval by Metadata", *Intl Conf on Soft Computing for Problem Solving*, **2017**. **SocProS**

22. A. Saboo, A. Sharma, T. Dash, "GASOM: Genetic Algorithm Assisted Architecture Learning in Self Organizing Maps", *International Conf on Neural Information Processing*, **2017**. **ICONIP**

23. P.N. Reddy, S.N. Dambekodi, T. Dash, "Towards Continuous Monitoring of Environment under Uncertainty: A Fuzzy Granular Decision Tree Approach", *ISEC Workshop on Development aspects of Intelligent Adaptive Systems workshop*, **2017**. **DIAS@ISEC**

24. T. Dash, T. Nayak, R.R. Swain, "Controlling Wall Following Robot Navigation Based on Gravitational Search and Feed Forward Neural Network", *International Conference on Perception and Machine Intelligence*, **2015**. **PerMIn**

25. T. Dash, S.K. Nayak, H.S. Behera, "Hybrid Gravitational Search and Particle Swarm Based Fuzzy MLP for Medical Data Classification", *International Conference on Computational Intelligence in Data Mining*, **2014**. **ICCIDM**

Additional collaborative publications in applied machine learning are on my Google Scholar profile.

Selected Preprints

1. T. Dash, G. Sachdeva, "p-adic Bi-Filtrations for Topological Machine Learning on Genomic Sequences", *arXiv*, **2026**. **arXiv**

2. E. Dadlani, T. Dash, D. Sahoo, "Investigating tumor-associated macrophages and their polarization in colorectal cancer using Boolean implication networks", *bioRxiv*, **2023**. **bioRxiv**

3. R. Khincha, S. Krishnan, T. Dash, L. Vig, A. Srinivasan, "Constructing and Evaluating an Explainable Model for COVID-19 Diagnosis from Chest X-rays", *arXiv*, **2020**. **arXiv**

Patents

1. Method and System for Iterative Knowledge Distillation for Neural Network Compression (Filed at Indian Patent Office, Appl. No. 202021055409, **2020**).

Selected Talks

1. Invited Talk: "Predicting gene expression using millions of yeast promoters reveals *cis*-regulatory logic", International Conference on Sustainable Healthcare from Biological Resources, BITS Pilani, Goa Campus, May 20-22, 2025; PdOC Research Showcase, Univ. of Cambridge, November 10, 2025.
2. Invited Talk: "Logically Explainable Deep Neural Networks", AI Club for Biomedicine, University of Cambridge, December 5, 2024.
3. Invited Talk: "Explainable Deep Learning" at Time-Series Forecasting Workshop, NIT Rourkela, India (online), September 28, 2024.
4. Poster Talk (with Ekta Dadlani): "Tumor Associated Macrophages and Colorectal Cancer: AI-assisted Predictive Modeling of Macrophage Polarization in Colorectal Cancer" at Structural & Functional Genomics 2023 Retreat, Moores Cancer Center, UC San Diego, May 5, 2023.
5. Invited Talk: "Graph Neural Networks: Concepts, Implementations and Application" at SAiDL-APPCAIR AI Symposium 2022, BITS Pilani, Goa Campus, India, October 9, 2022.
6. Invited Talk: "Deep Learning in a Human-in-the-Loop Setting" at IISER Pune, India, July 28, 2022.
7. Invited Talk: "Inclusion of domain-knowledge into GNNs using mode-directed inverse entailment" at The Intelligent Data Analysis Lab, Czech Technical University, Prague, June 30, 2022.
8. Invited Talk: "Human-in-the-Loop Machine Learning: Inclusion of Domain-Knowledge into Deep Neural Networks" at Davis Institute for Artificial Intelligence, USA, February 25, 2022.
9. Indo-German Spring School on Algorithms for Big Data, IIT Bombay, February 18-22, 2019 (also delivered a talk on: "Learning in the Presence of Expert Knowledge: A talk on Inductive Logic Programming").

Professional Services and Memberships

Memberships	IEEE and IEEE EMBS (Member, 2025-), ISCB (Member, 2025-), ACM (Member, 2019-), Trinity College (Postdoc Member, 2024, 2025; Alumnus, life-time), C2D3 Cambridge (2023-2025)
Editor	Discov. Artif. Intell., Adv. Comput. Intell., PLoS ONE
Peer review	J: JAIR, MLJ, IEEE (TPAMI, TCBB, TNNLS, TIE, CYB, TFS, CEM, IS), JCIM, Scientific Data, *and others* C: IJCAI, IJCAI-AI and Social Good, AAAI, CLeaR, ILP, DASFAA, IJCNN, ICANN
Co-organising	Cambridge AI Club for Biomedicine (2023-2025), ICCI 2017 (IITK)

Skills and Qualifications

GitHub	https://github.com/tirtharajdash
Languages	Spoken: English, Hindi, Odia; Programming: Python, MATLAB, Shell, C++
Contests	ACM ICPC (Regionals: 2012, 2013)
Qualifying exams	GATE (2012, 2015), UGC-NET (2014)
Certifications	Oracle Certified (SQL, DBA), 2012